



Optimal rotated staggered-grid finite-difference schemes for elastic wave modeling in TTI media



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ABSTRACT

The rotated staggered-grid finite-difference (RSFD) is an effective approach for numerical modeling to study the wavefield characteristics in tilted transversely isotropic (TTI) media. But it surfaces from serious numerical dispersion, which directly affects the modeling accuracy. In this paper, we propose two different optimal RSFD schemes based on the sampling approximation (SA) method and the least-squares (LS) method respectively to overcome this problem. We first briefly introduce the RSFD theory, based on which we respectively derive the SA-based RSFD scheme and the LS-based RSFD scheme. Then different forms of analysis are used to compare the SA-based RSFD scheme and the LS-based RSFD scheme with the conventional RSFD scheme, which is based on the Taylor-series expansion (TE) method. The contrast in numerical accuracy analysis verifies the greater accuracy of the two proposed optimal schemes, and indicates that these schemes can effectively widen the wavenumber range with great accuracy compared with the TE-based RSFD scheme. Further comparisons between these two optimal schemes show that at small wavenumbers, the SA-based RSFD scheme performs better, while at large wavenumbers, the LS-based RSFD scheme leads to a smaller error. Finally, the modeling results demonstrate that for the same operator length, the SA-based RSFD scheme and the LS-based RSFD scheme can achieve greater accuracy than the TE-based RSFD scheme, while for the same accuracy, the optimal schemes can adopt shorter difference operators to save computing time.

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1. Introduction

The anisotropy is widely existed in a subsurface medium, and the research on it can help provide fracture, lithology and fluid information, reflects geophysical or geological properties of subsurface structure, improve reservoir images, and further be used for better reservoir definition. Since the existence of tilted transversely isotropic (TTI) media is more general (Liu et al., 1998; Yan and Liu 2012), it is significant to perform numerical wavefield modeling in TTI media and analyze the feature of wavefield propagation. At present, the staggered-grid finite-difference (SFD) methods have been universally used in the seismic wave numerical modeling (e.g., Virieux 1984, 1986; O'Brien 2010; Moczo et al., 2014; Yang et al., 2014a; Zhang and Gao 2014). However, when we use standard staggered-grid finite-difference (SSFD) method (Virieux 1986) to solve the two-dimensional (2D), three-component wave equations for TTI media, some of the partial derivatives cannot be calculated directly, which needs an excursion or auxiliary grid. Additionally, in the standard staggered-grid format, the elastic modulus

is defined in both half grid points and whole grid points, which leads to an interpolation. If the variation of elastic modulus is serious, it would be easy to encounter the unstable calculation problem (Yan and Liu 2012).

Gold and Shapiro (1997) first proposed the rotated staggered-grid finite-difference (RSFD) method to overcome the instability problems for elastic wave modeling when the medium includes cracks or pores. Saenger et al. (2000), Saenger and Shapiro (2002), and Saenger and Bohlen (2004) also pointed out that RSFD can overcome the defect in the SSFD method, and further studied its stability and dispersion relation. Unlike the SSFD method, all the physical quantities of the RSFD are defined only at two different positions. Thus, all the elastic modulus can be defined at the same position such as the position of stress, and elastic modulus need not to be average, avoiding the interpolation (Chen et al., 2006; Bansal and Sen 2008). Therefore, the RSFD method can get more stable and convincing solutions in the simulation for TTI media (Yan and Liu 2012).

However, numerical dispersion error is unavoidable in every finite-difference method, and it is even more serious in RSFD than that in SSFD. The conventional high-order SFD scheme based on the Taylor-series expansion (TE) method can partly suppress the numerical dispersion over the range of small and medium frequency, but the difference coefficients calculated by this method cannot satisfy the

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requirement of accuracy in the whole seismic frequency zone, which leads to obvious numerical dispersion in wavefield propagation. To enhance the numerical solution accuracy, some scholars have proposed many ways to optimize the coefficients of the SFD operators, such as time-space domain dispersion-relation-based method (Liu and Sen 2011), simulated annealing algorithm (Zhang and Yao 2013), least-squares method (Yang et al., 2014a; Liu 2014) and so on. We (Yang et al., 2015) also proposed optimal implicit staggered-grid finite-difference schemes by using the sampling approximation. But, these optimization methods are only based on the standard staggered-grid format, and improve the numerical accuracy for the SSFD method. To decrease the numerical dispersion error in the RSFD method, it is extremely necessary to develop the optimization methods for the rotated staggered-grid format. We (Yang et al., 2014b) reported on the initial optimized formulas based on least-squares for rotated staggered-grid, but it could not consider the near zero-wavenumber constraint condition during the optimization process. In this paper, we explore the sampling approximation method and the least-squares method separately, and further develop them into the optimal FD schemes on rotated staggered-grid for elastic wave modeling in TTI media.

The structure of this paper is as follows: First, we propose the combination of the RSFD and the global optimization algorithms to derive the spatial difference coefficients, which are based on the sampling approximation (SA) method and the least-squares (LS) method, respectively. Then, the numerical accuracy of these RSFD schemes is analyzed according to the dispersion relation. We also use the optimal RSFD coefficients to solve the elastic wave equations for TTI media, and perform numerical modeling to examine the effectiveness of the new methods. Meanwhile, the modeling accuracy of the SA-based RSFD scheme and the LS-based RSFD scheme is compared with that of the TE-based RSFD scheme. Finally, we draw conclusions.

2. Theory of solving elastic wave equations with rotated staggered-grid finite-difference method

The elastic wave equations in 2D TTI media are (Winterstein 1990)

$$\frac{\partial v_x}{\partial t} = \frac{1}{\rho} \frac{\partial \sigma_{xx}}{\partial x} + \frac{1}{\rho} \frac{\partial \sigma_{xz}}{\partial z}, \quad (1a)$$

$$\frac{\partial v_y}{\partial t} = \frac{1}{\rho} \frac{\partial \sigma_{xy}}{\partial x} + \frac{1}{\rho} \frac{\partial \sigma_{yz}}{\partial z}, \quad (1b)$$

$$\frac{\partial v_z}{\partial t} = \frac{1}{\rho} \frac{\partial \sigma_{xz}}{\partial x} + \frac{1}{\rho} \frac{\partial \sigma_{zz}}{\partial z}, \quad (1c)$$

Besides, the transformation matrix R is expressed as (Liu et al., 1998)

$$R = \begin{bmatrix} \cos^2 \varphi & \sin^2 \varphi \cos^2 \theta & \sin^2 \varphi \sin^2 \theta & -\sin^2 \varphi \sin 2\theta & \sin 2\varphi \sin \theta & -\sin 2\varphi \cos \theta \\ \sin^2 \varphi & \cos^2 \varphi \cos^2 \theta & \cos^2 \varphi \sin^2 \theta & -\cos^2 \varphi \sin 2\theta & -\sin 2\varphi \sin \theta & \sin 2\varphi \cos \theta \\ 0 & \sin^2 \theta & \cos^2 \theta & \sin 2\theta & 0 & 0 \\ 0 & \frac{1}{2} \cos \varphi \sin 2\theta & -\frac{1}{2} \cos \varphi \sin 2\theta & \cos \varphi \cos 2\theta & \sin \varphi \cos \theta & \sin \varphi \sin \theta \\ 0 & -\frac{1}{2} \sin \varphi \sin 2\theta & \frac{1}{2} \sin \varphi \sin 2\theta & -\sin \varphi \cos 2\theta & \cos \varphi \cos \theta & \cos \varphi \sin \theta \\ \frac{1}{2} \sin 2\varphi & -\frac{1}{2} \sin 2\varphi \cos^2 \theta & -\frac{1}{2} \sin 2\varphi \sin^2 \theta & \frac{1}{2} \sin 2\varphi \sin 2\theta & -\cos 2\varphi \sin \theta & \cos 2\varphi \cos \theta \end{bmatrix}, \quad (4)$$

where φ and θ represent the azimuth angel and the tilt angel, respectively.

$$\frac{\partial \sigma_{xx}}{\partial t} = d_{11} \frac{\partial v_x}{\partial x} + d_{13} \frac{\partial v_z}{\partial z} + d_{14} \frac{\partial v_y}{\partial z} + d_{15} \left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \right) + d_{16} \frac{\partial v_y}{\partial x}, \quad (1d)$$

$$\frac{\partial \sigma_{zz}}{\partial t} = d_{13} \frac{\partial v_x}{\partial x} + d_{33} \frac{\partial v_z}{\partial z} + d_{34} \frac{\partial v_y}{\partial z} + d_{35} \left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \right) + d_{36} \frac{\partial v_y}{\partial x}, \quad (1e)$$

$$\frac{\partial \sigma_{yz}}{\partial t} = d_{14} \frac{\partial v_x}{\partial x} + d_{34} \frac{\partial v_z}{\partial z} + d_{44} \frac{\partial v_y}{\partial z} + d_{45} \left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \right) + d_{46} \frac{\partial v_y}{\partial x}, \quad (1f)$$

$$\frac{\partial \sigma_{xz}}{\partial t} = d_{15} \frac{\partial v_x}{\partial x} + d_{35} \frac{\partial v_z}{\partial z} + d_{45} \frac{\partial v_y}{\partial z} + d_{55} \left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \right) + d_{56} \frac{\partial v_y}{\partial x}, \quad (1g)$$

$$\frac{\partial \sigma_{xy}}{\partial t} = d_{16} \frac{\partial v_x}{\partial x} + d_{36} \frac{\partial v_z}{\partial z} + d_{46} \frac{\partial v_y}{\partial z} + d_{56} \left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \right) + d_{66} \frac{\partial v_y}{\partial x}, \quad (1h)$$

where t is the time, x and z are the space coordinates, ρ is the density, (v_x, v_y, v_z) is the velocity vector, $(\sigma_{xx}, \sigma_{zz})$ is the normal stress vector, $(\sigma_{xy}, \sigma_{xz}, \sigma_{yz})$ is the shear stress vector, and $D = [d_{ij}]_{6 \times 6}$ represents the elastic parameter matrix under the observation coordinate system.

D can be computed by another elastic parameter matrix $C = [c_{ij}]_{6 \times 6}$ which is under the natural coordinate system and the transformation matrix R (Winterstein 1990)

$$D = RCR^T. \quad (2)$$

The form of the elastic parameter matrix $C = [c_{ij}]_{6 \times 6}$ is

$$C = \begin{bmatrix} c_{11} & c_{12} & c_{13} & 0 & 0 & 0 \\ c_{12} & c_{11} & c_{13} & 0 & 0 & 0 \\ c_{13} & c_{13} & c_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & c_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & c_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & c_{66} \end{bmatrix}, \quad (3)$$

with $c_{66} = \frac{1}{2}(c_{11} - c_{12})$.

Generally, the temporal derivatives in the wave equations are calculated with the second-order finite-difference schemes, while the spatial derivatives are calculated with the high-order RSFD schemes. Fig. 1 shows the coordinate system of the RSFD method, where $\tilde{x}$ and $\tilde{z}$ are the space coordinates in the rotated staggered-grid coordinate system.

When the RSFD schemes are applied to the solution of the spatial derivatives, the partial derivatives in the rectangular coordinate system can be expressed as the partial derivatives in the rotated coordinate system as follows (Saenger and Bohlen 2004)

$$\frac{\partial}{\partial \tilde{x}} = \frac{\Delta r}{2\Delta x} \left(\frac{\partial}{\partial \tilde{z}} + \frac{\partial}{\partial \tilde{x}} \right), \quad (5a)$$

$$\frac{\partial}{\partial \tilde{z}} = \frac{\Delta r}{2\Delta z} \left(\frac{\partial}{\partial \tilde{z}} - \frac{\partial}{\partial \tilde{x}} \right). \quad (5b)$$

Δx and Δz are the space intervals along the x and z directions, respectively, while Δr is the space intervals along the $\tilde{x}$ and $\tilde{z}$ directions, which is also the length of the diagonal line, and $\Delta r = \sqrt{\Delta x^2 + \Delta z^2}$.

Using the plane wave theory, we make

$$u(x, z) = u_0 e^{i(k_x x + k_z z)}, \quad (6)$$

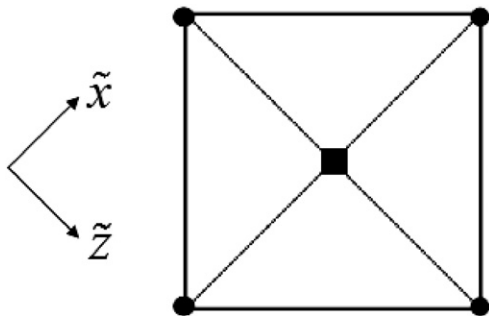
where u_0 is a constant, $i = \sqrt{-1}$, k_x and k_z are the x and z components of wavenumber, respectively.

In the rotated staggered-grid coordinate system, the partial derivatives along the $\tilde{x}$ and $\tilde{z}$ directions can be calculated with the (2M)th-order RSFD schemes (Yan and Liu 2012)

$$\frac{\partial}{\partial \tilde{x}} u(x, z, t) \approx \frac{1}{\Delta r} \sum_{m=1}^M a_m \left[u \left(x + \frac{(2m-1)}{2} \Delta x, z - \frac{(2m-1)}{2} \Delta z, t \right) - u \left(x - \frac{(2m-1)}{2} \Delta x, z + \frac{(2m-1)}{2} \Delta z, t \right) \right], \quad (7a)$$

$$\frac{\partial}{\partial \tilde{z}} u(x, z, t) \approx \frac{1}{\Delta r} \sum_{m=1}^M a_m \left[u \left(x + \frac{(2m-1)}{2} \Delta x, z + \frac{(2m-1)}{2} \Delta z, t \right) - u \left(x - \frac{(2m-1)}{2} \Delta x, z - \frac{(2m-1)}{2} \Delta z, t \right) \right], \quad (7b)$$

where a_m ($m = 1, 2, \dots, M$) are the coefficients of the RSFD schemes.



● (i, j) ρ, v_x, v_y, v_z

■ $(i+1/2, j+1/2)$

$c_{ij}, \sigma_{xx}, \sigma_{yy}, \sigma_{zz}, \sigma_{yz}, \sigma_{xz}, \sigma_{xy}$

Fig. 1. Stencil of discretization scheme in 2D rotated staggered-grid coordinate system (Saenger et al., 2000).

Taking the partial derivative along x component as an example, we substitute Eqs. (6) and (7a) into Eq. (5a), and obtain

$$\frac{\Delta x k_x}{2} \approx \sum_{m=1}^M a_m \sin \left(\frac{2m-1}{2} k_x \Delta x \right) \cos \left(\frac{2m-1}{2} k_z \Delta z \right). \quad (8)$$

We suppose

$$k_x = k_z \quad (9)$$

$$\Delta x = \Delta z, \quad (10)$$

and let

$$k_x \Delta x = k_z \Delta z = \beta, \quad (11)$$

$$f_m(\beta) = \sin[(2m-1)\beta], \quad (12)$$

where $\beta \in [0, u]$. Here u is a constant, and $u \in (0, \pi/2]$, because $k_x \Delta x$ and $k_z \Delta z$ equal $\pi/2$ at the Nyquist frequency.

Substitute Eqs. (11) and (12) into (8), and obtain

$$\beta \approx \sum_{m=1}^M a_m f_m(\beta), \quad (13)$$

which shows the dispersion relation of the RSFD method.

3. Derivation of the rotated staggered-grid finite-difference schemes

3.1. Rotated staggered-grid finite-difference scheme based on Taylor-series expansion method

The dispersion relation shown in Eq. (13) is crucial for the derivation of the RSFD schemes, based on which we first briefly introduce the derivation of the conventional TE-based RSFD scheme.

We first apply the TE method to $f_m(\beta)$, and it can be expanded at $\beta = 0$ as

$$f_m(\beta) = \sum_{n=1}^{+\infty} \frac{(2m-1)^{2n-1} (-1)^{n-1} \beta^n}{(2n-1)!}. \quad (14)$$

Substituting Eqs. (14) to (13), and comparing the coefficients of β , we get the following equation (Liu and Sen 2009)

$$\frac{(-1)^{n-1}}{(2n-1)!} \sum_{m=1}^M [(2m-1)^{2n-1}] a_m = \begin{cases} 1, & n = 1 \\ 0, & n = 2, 3, \dots, M \end{cases}, \quad (15)$$

which can be also written as the following M linear equations (Pei 2004; Liu and Sen 2009)

$$\begin{bmatrix} 1 & 3^1 & \dots & (2M-1)^1 \\ 1 & 3^3 & \dots & (2M-1)^3 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 3^{2M-1} & \dots & (2M-1)^{2M-1} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_M \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}. \quad (16)$$

After solving the linear Eq. (16) by Gaussian elimination method, we can get the TE-based RSFD coefficients, which are listed in Table 1.

3.2. Optimal rotated staggered-grid finite-difference scheme based on sampling approximation method

In this section, we propose optimal RSFD scheme based on SA method.

From Eq. (13), we can see that the left part is the reserved item of the spatial derivative, while the right part is the reserved item of the RSFD scheme. Since the spatial derivatives are calculated with the difference formulas, this would lead to an error between the left

Table 1

The TE-based RSFD coefficients for the first-order spatial derivatives.

	$M = 2$	$M = 3$	$M = 4$	$M = 5$	$M = 6$
a_1	1.125000e + 0	1.171875e + 0	1.196289e + 0	1.211243e + 0	1.221336e + 0
a_2	−4.166667e − 2	−6.510417e − 2	−7.975260e − 2	−8.972168e − 2	−9.693146e − 2
a_3		4.687500e − 2	9.570313e − 3	1.384277e − 2	1.744766e − 2
a_4			−6.975446e − 4	−1.765660e − 3	−2.967290e − 3
a_5				1.186795e − 4	3.590054e − 4
a_6					−2.184781e − 5
	$M = 7$	$M = 8$	$M = 9$	$M = 10$	$M = 11$
a_1	1.228606e + 0	1.234091e + 0	1.238376e + 0	1.241816e + 0	1.244638e + 0
a_2	−1.023839e − 1	−1.066498e − 1	−1.100779e − 1	−1.128924e − 1	−1.152443e − 1
a_3	2.047677e − 2	2.303637e − 2	2.521784e − 2	2.709417e − 2	2.872242e − 2
a_4	−4.178933e − 3	−5.342386e − 3	−6.433123e − 3	−7.443453e − 3	−8.373884e − 3
a_5	6.894535e − 4	1.077271e − 3	1.496785e − 3	1.929784e − 3	2.363985e − 3
a_6	−7.692250e − 5	−1.664189e − 4	−2.862801e − 4	−4.306130e − 4	−5.934385e − 4
a_7	4.236515e − 6	1.702171e − 5	4.099395e − 5	7.707717e − 5	1.249670e − 4
a_8		−8.523464e − 7	−3.848877e − 6	−1.021650e − 5	−2.085870e − 5
a_9			1.762665e − 7	8.837806e − 7	2.564127e − 6
a_{10}				−3.723759e − 8	−2.052722e − 7
a_{11}					8.001648e − 9

part and the right part of Eq. (13), and the error accumulates over the variation range of β . In order to control this error over $[0, u]$, we take the SA method, which makes the right part of Eq. (13) approach the left part, and further the RSFD scheme approximate the spatial derivatives.

We first take M sampling points evenly over the range of β . This uniform sampling can give fair consideration to every local detail over $[0, u]$. At these sampling points, which can be denoted by $\beta(1), \beta(2), \dots, \beta(M)$, we make the right part of Eq. (13) equal the left part.

When $\beta = \beta(1) = 0$, the equality of Eq. (13) can be unconditional. Thus, we use the near zero-wavenumber constraint condition

$$\lim_{\beta \rightarrow 0} \sum_{m=1}^M a_m f_m(\beta) / \beta = 1, \quad (17)$$

which provides the relation between a_1 and $a_m (m = 2, \dots, M)$ (Liu 2014)

$$a_1 = 1 - \sum_{m=2}^M a_m (2m-1). \quad (18)$$

Substituting Eq. (18) into (13), we obtain

$$\sum_{m=2}^M [f_m(\beta) - (2m-1)f_1(\beta)] a_m \approx \beta - f_1(\beta). \quad (19)$$

Subsequently, for $\beta = \beta(2), \beta = \beta(3), \dots, \beta = \beta(M)$, we use these $(M-1)$ sampling points to construct linear equations

$$\begin{bmatrix} g_2(\beta(2)) & g_3(\beta(2)) & \cdots & g_M(\beta(2)) \\ g_2(\beta(3)) & g_3(\beta(3)) & \cdots & g_M(\beta(3)) \\ \vdots & \vdots & \ddots & \vdots \\ g_2(\beta(M)) & g_3(\beta(M)) & \cdots & g_M(\beta(M)) \end{bmatrix} \times \begin{bmatrix} a_2 \\ a_3 \\ \vdots \\ a_M \end{bmatrix} = \begin{bmatrix} \beta(2) - f_1(\beta(2)) \\ \beta(3) - f_1(\beta(3)) \\ \vdots \\ \beta(M) - f_1(\beta(M)) \end{bmatrix}, \quad (20)$$

where $g_m(\beta) = f_m(\beta) - (2m-1)f_1(\beta)$.

$a_m (m = 2, 3, \dots, M)$ can be obtained by solving Eq. (20) with Gaussian elimination method, and a_1 is calculated using Eq. (18). The SA-based RSFD coefficients are listed in Table 2.

The process of uniform sampling over the whole range of β and making both the sides of Eq. (13) equal at the sampling points, results in the RSFD scheme's approximation to the spatial derivatives. Thus, this

Table 2The SA-based RSFD coefficients for the first-order spatial derivatives ($u = 1.10$).

	$M = 2$	$M = 3$	$M = 4$	$M = 5$	$M = 6$
a_1	1.221228e + 0	1.227062e + 0	1.234996e + 0	1.241258e + 0	1.245920e + 0
a_2	−7.374268e − 2	−9.641949e − 2	−1.059046e − 1	−1.117645e − 1	−1.159598e − 1
a_3		1.243930e − 2	2.056187e − 2	2.536470e − 2	2.867786e − 2
a_4			−2.870219e − 3	−5.669177e − 3	−7.822755e − 3
a_5				7.662654e − 4	1.753131e − 3
a_6					−2.226460e − 4
	$M = 7$	$M = 8$	$M = 9$	$M = 10$	$M = 11$
a_1	1.249453e + 0	1.252200e + 0	1.254390e + 0	1.256173e + 0	1.257652e + 0
a_2	−1.191243e − 1	−1.215958e − 1	−1.235782e − 1	−1.252030e − 1	−1.265585e − 1
a_3	3.118437e − 2	3.316826e − 2	3.478415e − 2	3.612810e − 2	3.726438e − 2
a_4	−9.524111e − 3	−1.091945e − 2	−1.209140e − 2	−1.309224e − 2	−1.395795e − 2
a_5	2.681605e − 3	3.512563e − 3	4.253120e − 3	4.914807e − 3	5.508351e − 3
a_6	−5.778290e − 4	−9.709608e − 4	−1.364307e − 3	−1.743967e − 3	−2.104565e − 3
a_7	6.837094e − 5	1.982285e − 4	3.630081e − 4	5.452040e − 4	7.347689e − 4
a_8		−2.182960e − 5	−6.987798e − 5	−1.384972e − 4	−2.215383e − 4
a_9			7.174235e − 6	2.511851e − 5	5.356334e − 5
a_{10}				−2.410968e − 6	−9.162103e − 6
a_{11}					8.247187e − 7

method is called the SA method. The exact equivalence of Eq. (19) at the sampling points controls the numerical errors in the neighborhood of these points, while the constraint condition near zero-wavenumber improves the accuracy at small frequency or wavenumbers.

3.3. Optimal rotated staggered-grid finite-difference scheme based on least squares method

We also start the derivation of the LS-based RSFD scheme based on Eq. (13). From Eq. (13), we can see that the difference between the left part and the right part shows the error introduced by the RSFD scheme, and the integral of this error over $[0, u]$ measures the accumulated error over the range of β . We define an integral expression (e.g., Yang et al., 2014b; Liu 2014)

$$I(a_1, a_2, \dots, a_M) = \int_0^u \left[\beta - \sum_{m=1}^M a_m f_m(\beta) \right]^2 d\beta, \quad (21)$$

where $I(a_1, a_2, \dots, a_M)$ is also the quadratic function about a_m . We expect this integral to decrease to a minimum value, so that the accumulated error is small. Therefore, we apply the LS method here, and according to the necessary condition of extremum, we get

$$\frac{\partial I(a_1, a_2, \dots, a_M)}{\partial a_m} = 0. \quad (22)$$

In order to enhance the precision near zero-wavenumber, we adopt the constraint condition Eq. (17) again and substitute Eq. (18) into Eq. (21). Thus, the LS method shown in Eq. (22) provides the following $(M-1)$ linear equations

$$\begin{bmatrix} (g_2, g_2) & (g_3, g_2) & \cdots & (g_M, g_2) \\ (g_2, g_3) & (g_3, g_3) & \cdots & (g_M, g_3) \\ \vdots & \vdots & \ddots & \vdots \\ (g_2, g_M) & (g_3, g_M) & \cdots & (g_M, g_M) \end{bmatrix} \begin{bmatrix} a_2 \\ a_3 \\ \vdots \\ a_M \end{bmatrix} = \begin{bmatrix} (t, g_2) \\ (t, g_3) \\ \vdots \\ (t, g_M) \end{bmatrix} \quad (23)$$

where $(g_i, g_j) = \int_0^u g_i(\beta) \cdot g_j(\beta) d\beta$, $(t, g_j) = \int_0^u t(\beta) \cdot g_j(\beta) d\beta$, $i, j = 2, 3, \dots, M$; $t(\beta) = \beta - f_1(\beta)$, and $g_i(\beta)$ has the same expression as that in Section 3.2.

The LS-based RSFD coefficients can also be obtained by solving Eq. (23) with Gaussian elimination method together with Eq. (18). And these coefficients are listed in Table 3.

Table 3
The LS-based RSFD coefficients for the first-order spatial derivatives ($u = 1.10$).

	$M = 2$	$M = 3$	$M = 4$	$M = 5$	$M = 6$
a_1	1.190325e + 0	1.220278e + 0	1.234822e + 0	1.242970e + 0	1.248264e + 0
a_2	-6.344150e - 2	-9.501921e - 2	-1.063796e - 1	-1.134665e - 1	-1.180869e - 1
a_3		1.295599e - 2	2.192849e - 2	2.685699e - 2	3.040395e - 2
a_4			-3.618001e - 3	-6.762350e - 3	-8.998313e - 3
a_5				1.164592e - 3	2.381875e - 3
a_6					-4.065231e - 4
	$M = 7$	$M = 8$	$M = 9$	$M = 10$	$M = 11$
a_1	1.251971e + 0	1.254716e + 0	1.256831e + 0	1.258511e + 0	1.259879e + 0
a_2	-1.213927e - 1	-1.238695e - 1	-1.257985e - 1	-1.273438e - 1	-1.286104e - 1
a_3	3.300418e - 2	3.501448e - 2	3.661340e - 2	3.791767e - 2	3.900231e - 2
a_4	-1.079739e - 2	-1.224628e - 2	-1.344395e - 2	-1.444897e - 2	-1.530476e - 2
a_5	3.418801e - 3	4.336199e - 3	5.135864e - 3	5.837516e - 3	6.455769e - 3
a_6	-9.040881e - 4	-1.391068e - 3	-1.859480e - 3	-2.296991e - 3	-2.702282e - 3
a_7	1.495573e - 4	3.597311e - 4	5.902296e - 4	8.292283e - 4	1.066726e - 3
a_8		-5.708811e - 5	-1.477947e - 4	-2.574460e - 4	-3.791922e - 4
a_9			2.239569e - 5	6.213162e - 5	1.144707e - 4
a_{10}				-8.973453e - 6	-2.657198e - 5
a_{11}					3.656647e - 6

4. Numerical accuracy analysis

We have derived the TE-based RSFD scheme, the SA-based RSFD scheme and the LS-based RSFD scheme, and have computed their coefficients, respectively. For the same RSFD method, when different series of M and u are applied, we can get different series of coefficients a_m . At the same time, when different series of a_m are applied in the RSFD schemes to the solution of the spatial derivatives, the numerical solution accuracy is also different. Consequently, the error ε introduced by the RSFD method can be expressed as the function of M and u

$$\varepsilon = \sqrt{\frac{1}{u} \int_0^u [\delta_M(\beta)]^2 d\beta}, \quad (24)$$

where $\delta_M(\beta)$ is defined according to the dispersion relation shown in Eq. (13)

$$\delta_M(\beta) = \sum_{m=1}^M a_m f_m(\beta) / \beta - 1. \quad (25)$$

If $\delta_M(\beta) = 0$, no numerical dispersion exists. If $\delta_M(\beta)$ is far from 0, large numerical dispersion will occur.

Fig. 2 illustrates the variations of $\log_{10}\varepsilon$ with u for different M by the TE-based RSFD scheme, the SA-based RSFD scheme and the LS-based RSFD scheme, respectively. Meanwhile, we design Table 4, which shows the detailed u values for different ε and different M of these three RSFD schemes. The similarities in Fig. 2(a), (b), and (c) indicate that

- When M is given, the numerical errors introduced by these three RSFD schemes increase with the increase of u .
- When u is given, ε decreases with the increase of M .
- When ε is given, u increases with the increase of M , which means for the specific accuracy, the schemes with long difference operators can widen the wavenumber range.

However, when comparing Fig. 2(a), (b), and (c), as well as the detailed u values for different schemes listed in Table 4, we can see that

- For a given M and a specific u , the numerical errors introduced by the SA-based RSFD scheme and the LS-based RSFD scheme are much smaller than that of the TE-based RSFD scheme. For example, when $M = 11$, and $u = 1.2$, the error introduced by the TE-based RSFD scheme is 1.0×10^{-3} , while for the SA-based RSFD scheme and

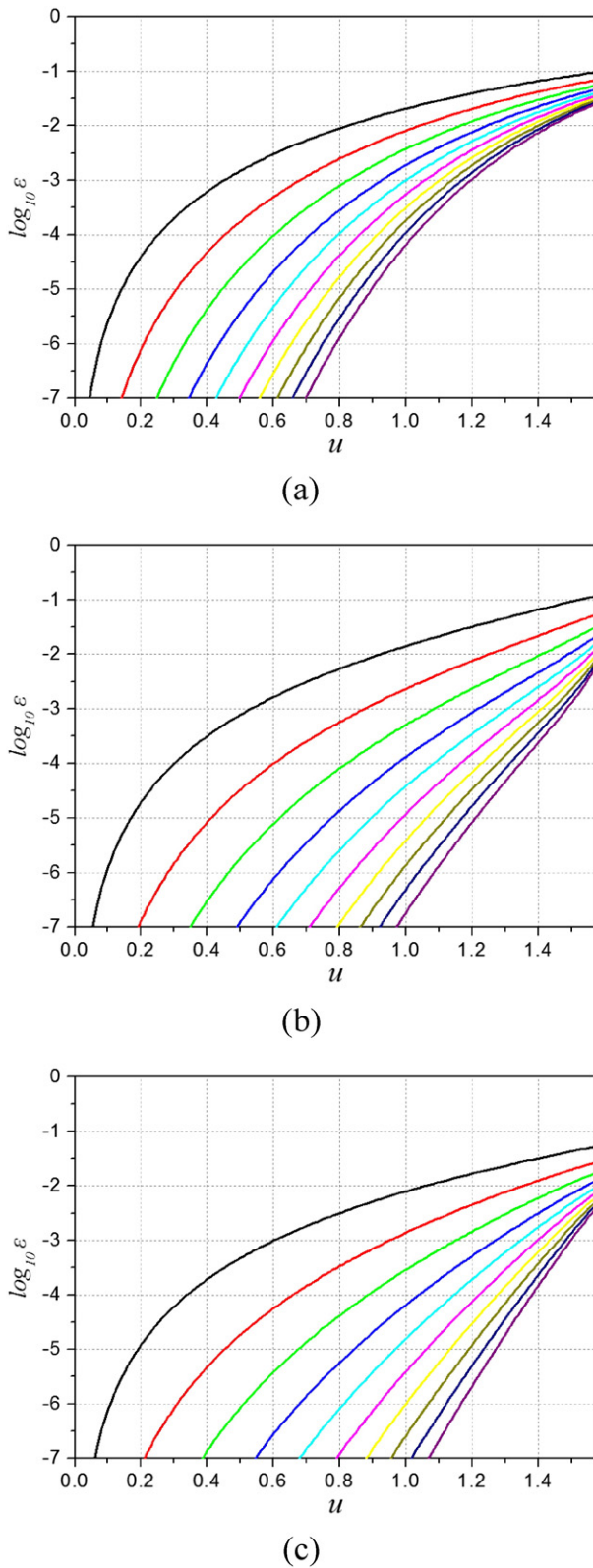


Fig. 2. Variations of $\log_{10}\varepsilon$ with u for different M by (a) the TE-based RSFD scheme, (b) the SA-based RSFD scheme and (c) the LS-based RSFD scheme. From left to right, curves are from $M = 2, 3, \dots, 11$, respectively.

the LS-based RSFD scheme, both of their errors are smaller than 1.0×10^{-5} . The comparison indicates that the optimal methods can decrease the numerical error effectively.

- For a specific ε and a given M , u values of the SA-based RSFD scheme

Table 4

u values for different ε and different M by the TE-based RSFD scheme, the SA-based RSFD scheme and the LS-based RSFD scheme.

	u					
	$\varepsilon = 10^{-6}$			$\varepsilon = 10^{-5}$		
	TE	SA	LS	TE	SA	LS
$M = 2$	0.08	0.10	0.11	0.14	0.17	0.19
$M = 3$	0.21	0.28	0.31	0.31	0.41	0.45
$M = 4$	0.33	0.46	0.51	0.45	0.62	0.67
$M = 5$	0.44	0.62	0.68	0.55	0.78	0.84
$M = 6$	0.52	0.74	0.81	0.64	0.90	0.97
$M = 7$	0.59	0.84	0.92	0.71	0.99	1.06
$M = 8$	0.65	0.92	1.00	0.77	1.06	1.13
$M = 9$	0.70	0.98	1.07	0.82	1.12	1.19
$M = 10$	0.75	1.04	1.12	0.86	1.17	1.23
$M = 11$	0.79	1.08	1.17	0.90	1.21	1.27

	$\varepsilon = 10^{-4}$			$\varepsilon = 10^{-3}$		
	TE	SA	LS	TE	SA	LS
$M = 2$	0.25	0.30	0.34	0.45	0.53	0.60
$M = 3$	0.45	0.60	0.66	0.68	0.87	0.95
$M = 4$	0.60	0.82	0.89	0.83	1.09	1.15
$M = 5$	0.71	0.97	1.04	0.93	1.22	1.27
$M = 6$	0.79	1.08	1.15	1.00	1.31	1.35
$M = 7$	0.86	1.17	1.22	1.06	1.37	1.40
$M = 8$	0.91	1.23	1.28	1.10	1.41	1.43
$M = 9$	0.96	1.28	1.32	1.14	1.44	1.46
$M = 10$	1.00	1.32	1.35	1.17	1.47	1.48
$M = 11$	1.03	1.35	1.38	1.20	1.49	1.50

and the LS-based RSFD scheme are much larger than that of the TE-based RSFD scheme. For example, when $\varepsilon = 1.0 \times 10^{-5}$, and $M = 8$, u value of the TE-based RSFD scheme equals 0.77, while those of the SA-based RSFD scheme and the LS-based RSFD scheme equal 1.06 and 1.13, which expand the wavenumber range by nearly 1/3. This comparison shows that the newly proposed methods can widen the wavenumber range with great accuracy.

- For specific ε and u values, the operator length of the SA-based RSFD scheme and the LS-based RSFD scheme is shorter than that of the TE-based RSFD scheme. For example, for a specific β range $[0, 1.20]$, the errors calculated by Eq. (24) for the 22th-order TE-based RSFD scheme, the 10th-order SA-based RSFD scheme and the 10th-order LS-based RSFD scheme are all about 1.0×10^{-3} . Furthermore, Fig. 3 displays this comparison. Although these three schemes lead to the similar errors within $[0, 1.20]$ the SA-based RSFD scheme and the LS-based RSFD scheme save the operator length dramatically.

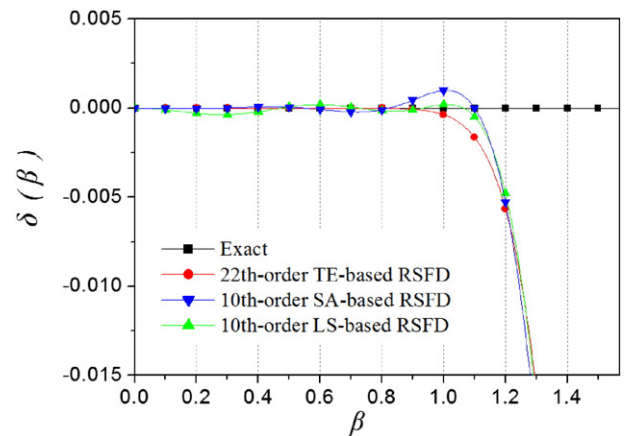


Fig. 3. Plot of the dispersion curves by the 22th-order TE-based RSFD scheme, the 10th-order SA-based RSFD scheme and the 10th-order LS-based RSFD scheme, which have the similar errors within $[0, 1.20]$.

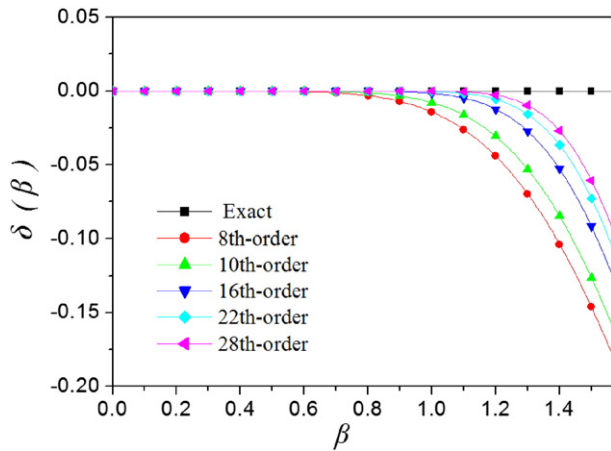


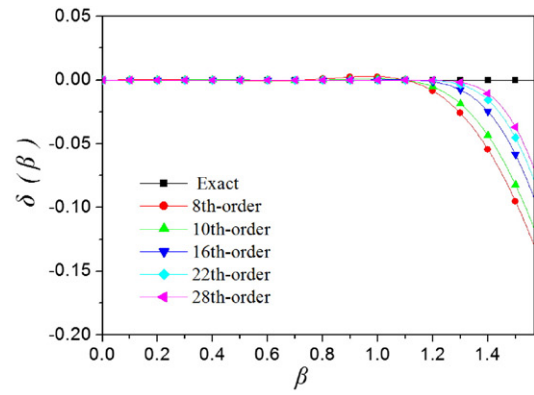
Fig. 4. Plot of numerical dispersion curves for different operator length by the TE-based RSFD scheme.

Figs. 4 and 5 display the numerical dispersion curves by the TE-based RSFD scheme, the SA-based RSFD scheme, and the LS-based RSFD scheme for different operator length. From these figures, we can see that the dispersion increases with the increase of the wavenumber, while decreases with the increase of the operator length, generally.

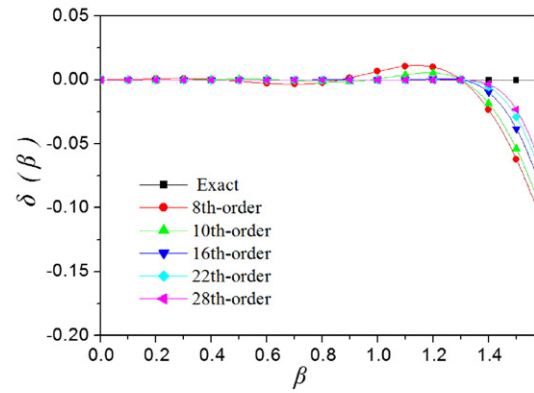
Since the computation of the coefficients of the SA-based RSFD and LS-based RSFD schemes is dependant on u value, we also display dispersion curves with different u values by these two schemes in Fig. 5. The comparison of Fig. 5(a) and (b) or (c) and (d) tells us that when we adopt the SA-based RSFD or LS-based RSFD scheme to solve the spatial derivatives, the numerical accuracy relies on the values of u . Generally, the RSFD schemes with smaller u values can achieve great accuracy at small and medium wavenumbers, while those with larger u values improve the numerical solution accuracy at large wavenumbers, and widen the wavenumber range with great accuracy to some extent. However, a larger u sometimes means a sacrifice of accuracy at some local details, and makes the dispersion curves fluctuate with instability within the variation range of β . Fig. 6 displays the dispersion curves by the SA-based RSFD and LS-based RSFD schemes for different u values under the same scale, which can further demonstrate the conclusions above. Meanwhile, when we make comparisons between the SA-based RSFD and LS-based RSFD schemes with the same u value in Fig. 6, we can find that at small wavenumbers, the SA-based RSFD schemes tend to get smaller numerical errors than the LS-based RSFD schemes, while at large wavenumbers, the LS-based RSFD schemes suppress numerical dispersion more effectively than the SA-based RSFD schemes.

When comparing Figs. 4, 5(a) and (c), we can see that at small wavenumbers, the numerical dispersion can be effectively suppressed by all of these three RSFD schemes. However, over the range of large wavenumbers, the dispersion curves by the TE-based RSFD scheme greatly go off the zero-dispersion curve, while the SA-based and LS-based RSFD schemes keep the relative superiority, and widen the wavenumber range with great accuracy. Fig. 7 further displays these differences and obviously shows that the SA-based RSFD and LS-based RSFD schemes do not only achieve great accuracy at small wavenumbers as the TE-based RSFD scheme does, but also greatly improve the numerical accuracy at large wavenumbers. The contrasts among these three RSFD methods demonstrate that the SA-based RSFD and LS-based RSFD schemes achieve greater accuracy than the TE-based RSFD scheme.

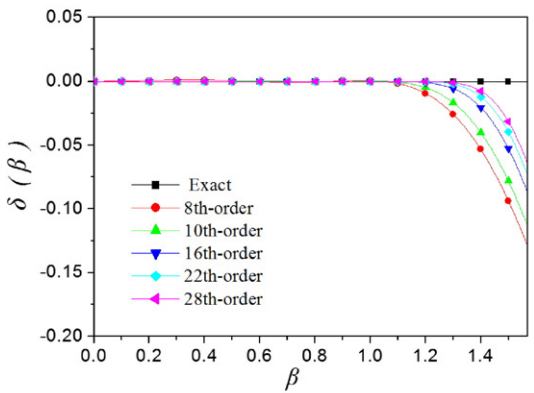
The numerical accuracy analysis demonstrates that for the same operator length, the SA-based RSFD scheme and the LS-based RSFD



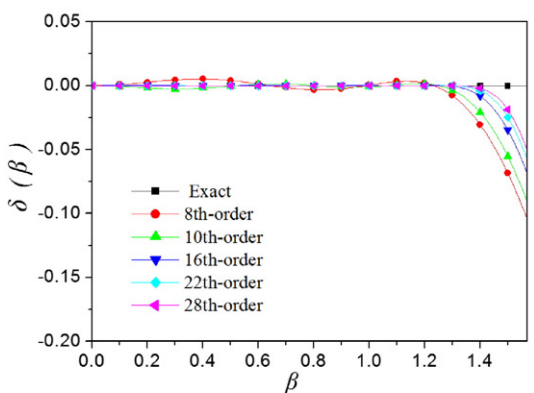
(a)



(b)



(c)



(d)

Fig. 5. Plot of numerical dispersion curves for different operator length by (a) the SA-based RSFD scheme, when $u = 1.10$, (b) the SA-based RSFD scheme, when $u = 1.30$, (c) the LS-based RSFD scheme, when $u = 1.10$, (d) the LS-based RSFD scheme, when $u = 1.30$.

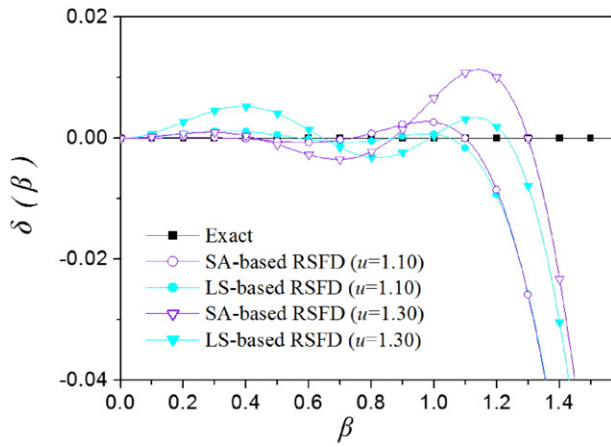


Fig. 6. Plot of numerical dispersion curves by the 8th-order SA-based RSFD and LS-based RSFD schemes, when $u = 1.10$ and 1.30 , respectively.

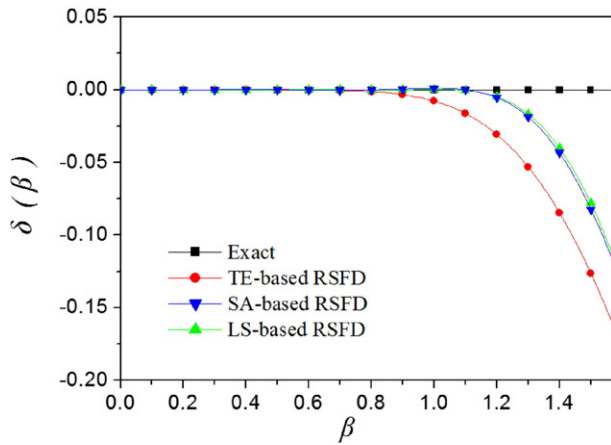


Fig. 7. Plot of numerical dispersion curves by the different RSFD schemes for the 10th-order accuracy, when $u = 1.10$.

scheme can decrease the numerical errors, globally improve the accuracy, and widen the range of wavenumber with great accuracy compared with the TE-based RSFD scheme. Meanwhile, for the same numerical accuracy, the SA-based RSFD scheme and the LS-based RSFD scheme save the operator length, which results in a saving in computational cost. Since the only difference among the optimal methods and the TE-based RSFD method lies in the calculation of the difference coefficients, whose computing time can be approximately the same, the computational costs for these methods actually depend on the operator length in the FD scheme. Hence, the optimal methods can reduce the operator length, save the computing time and further improve the computing efficiency. In addition, the difference between the SA-based RSFD scheme and the LS-based RSFD scheme shows that the former achieves relatively greater accuracy at small wavenumbers, while the latter leads to superiority over the range of large wavenumbers.

5. Numerical examples

5.1. Numerical modeling in the homogeneous medium

We first perform numerical modeling in a 2D homogeneous TTI medium to validate the modeling results by the TE-based RSFD scheme, the SA-based RSFD scheme and the LS-based RSFD scheme. The grid dimension is 2000 m, with a grid interval of 5 m in both x and z directions. The elastic parameters (c_{ij}) together with the density ρ are listed in Table 5, where P-wave velocity and S-wave velocity are 1732 m/s and 800 m/s, respectively. The azimuth angle is 45° , and the tilt angle is 60° . As the source function, a Ricker wavelet with the dominant frequency of 20 Hz is located in the center. The record-length is 1.8 s with a time step of 0.001 s.

We show some modeling results by the TE-based RSFD scheme, the SA-based RSFD scheme and the LS-based RSFD scheme with the 2nd-order accuracy in time and the 8th-order RSFD accuracy in space. Only x component is displayed for simplicity. Fig. 8 are the snapshots computed by these three methods at the moment of 0.5 s, and Fig. 9 are the seismograms. In these figures, we can clearly see qP wave, which has the largest propagation velocity, fast shear wave qS1 and slow shear wave qS2, whose existence means the phenomenon of shear wave splitting. Additionally, the asymmetric distorted lineups in Fig. 9 show the anisotropy in TTI media. When we compare the snapshots and seismograms by different methods, we find that in Figs. 8(a) and 9(a), numerical dispersion is obvious, while in contrast the lineups and wavefield are clear in Figs. 8(b), (c), 9(b) and (c). These modeling results demonstrate that the newly proposed SA-based RSFD and LS-based RSFD schemes perform better in suppressing numerical dispersion and achieve greater modeling precision than the TE-based RSFD scheme.

5.2. Numerical modeling in the complex medium

To make further comparisons among the TE-based RSFD scheme, the SA-based RSFD scheme and the LS-based RSFD scheme, we also implement elastic wave modeling in a local region of the modified 2D BP 2007 TTI model (Shah 2010). We use the Thomson anisotropic parameters (P-wave anisotropic parameter (η), variation coefficient (σ) and tilt angle (θ)) to describe this TTI model, whose relation with the elastic parameters (c_{ij}) in VTI media can be referred to the paper of Thomsen (1986).

Fig. 10 displays the parameter values for P-wave velocity (v_p), η , σ , and θ in this modeling region with the size of 15,000 m $\times$ 16,000 m. The ratio of S-wave velocity (v_s) and v_p is fixed at 1/1.7 for the same grid point, and that of S-wave anisotropic parameter (γ) and η is 1/1.15. The density and the azimuth angle are constants of 3200 kg/m³ and 45° , respectively. We place a Ricker wavelet with the dominant frequency of 20 Hz at (7500 m, 230 m) to generate vibration. The spatial interval and time step are 10 m and 0.0005 s, respectively, and the record length is 4.5 s.

Figs. 11, 12 and 13 show the magnification of some details in the wavefield snapshots computed by the TE-based RSFD scheme, the SA-based RSFD scheme and the LS-based RSFD scheme at 2.85 s, respectively. Meanwhile, Figs. 14, 15, and 16 display the seismograms by these three schemes. Both the 10th-order accuracy and the 18th-order accuracy are applied in the numerical modeling, and we only display x component here for simplicity. Because of the complexity of the

Table 5
The elastic parameters of the homogeneous elastic TTI media in the normal axis.

$c_{11}/(10^9\text{N/m}^2)$	$c_{13}/(10^9\text{N/m}^2)$	$c_{33}/(10^9\text{N/m}^2)$	$c_{44}/(10^9\text{N/m}^2)$	$c_{66}/(10^9\text{N/m}^2)$	$\rho/(10^3\text{ kg/m}^3)$
10.123	3.093	8.996	1.925	3.850	2.8

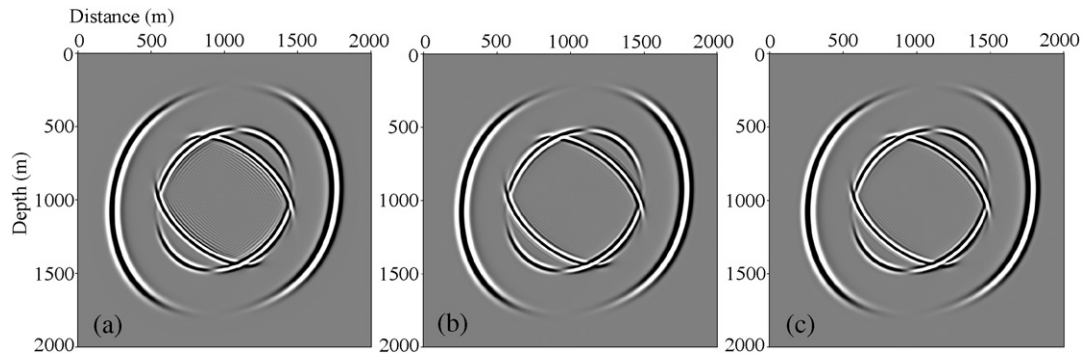


Fig. 8. Snapshots computed by the 8th-order scheme: (a) the TE-based RSFD method, (b) the SA-based RSFD method, (c) the LS-based RSFD method.

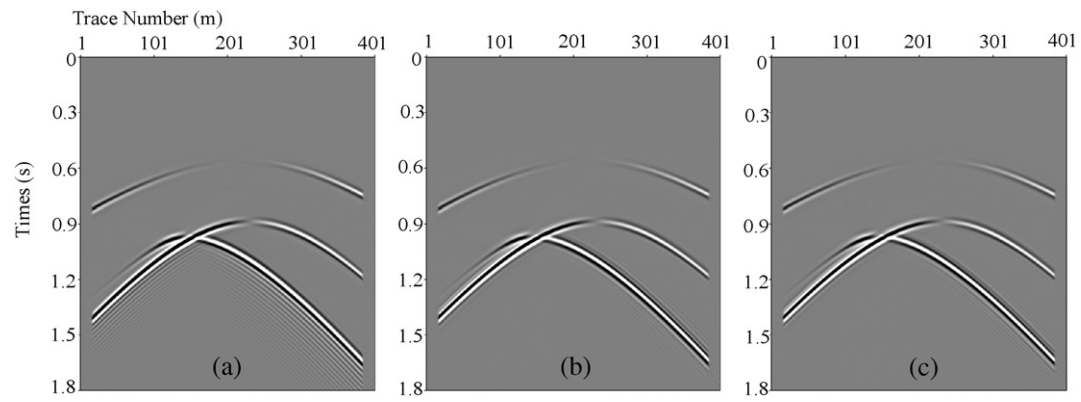


Fig. 9. Seismograms computed by the 8th-order scheme: (a) the TE-based RSFD method, (b) the SA-based RSFD method, (c) the LS-based RSFD method.

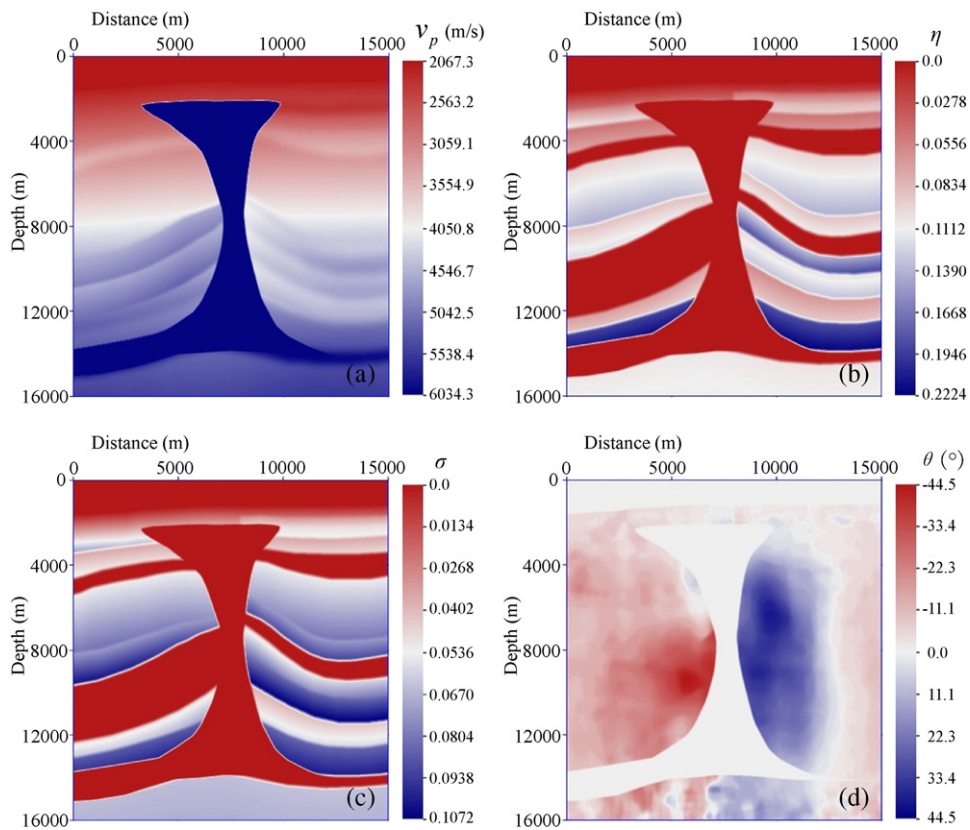


Fig. 10. Partial region of the modified 2D BP 2007 TTI model: (a) v_p , (b) η , (c) σ , (d) θ .

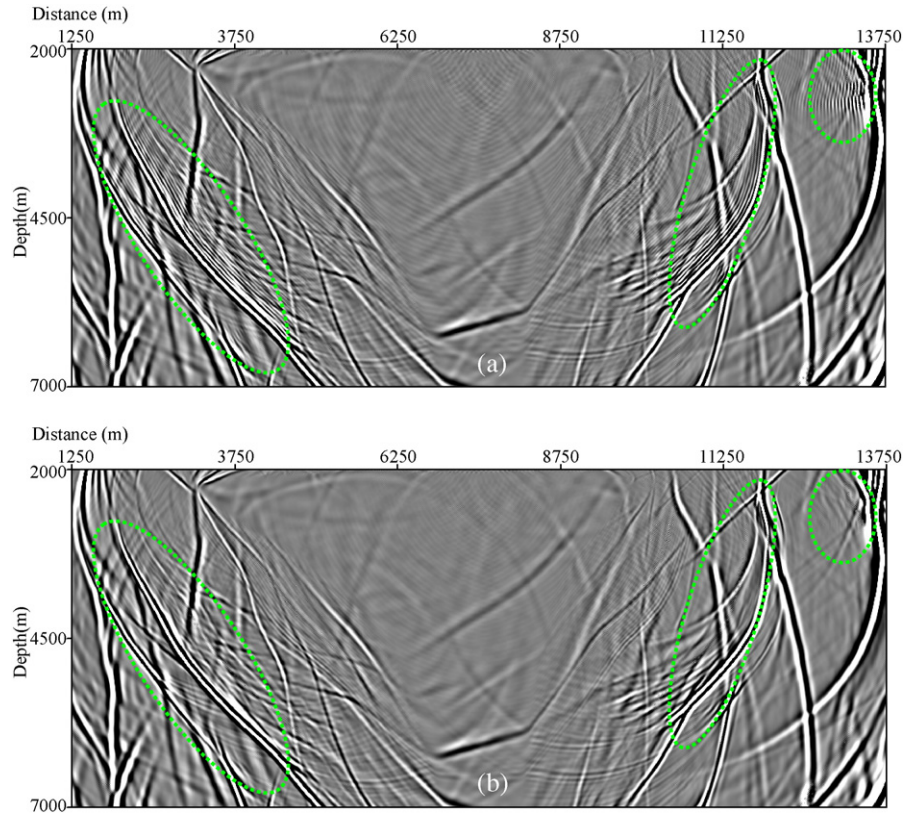


Fig. 11. Snapshots computed by the TE-based RSFD scheme: (a) the 10th-order accuracy, (b) the 18th-order accuracy.

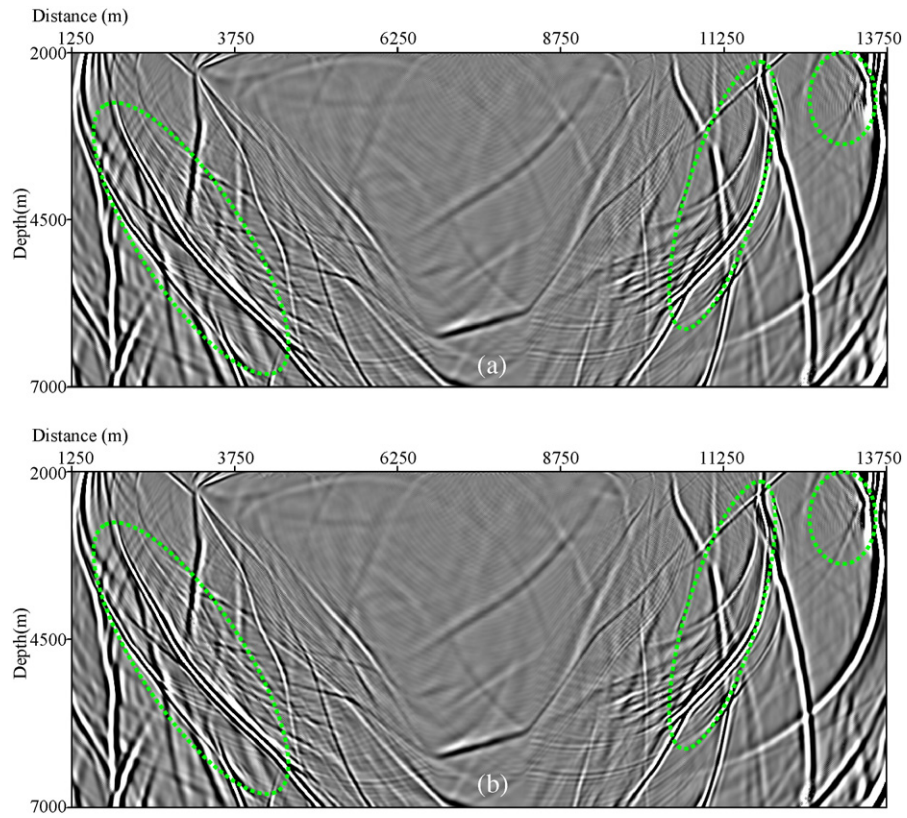


Fig. 12. Snapshots computed by the SA-based RSFD scheme: (a) the 10th-order accuracy, (b) the 18th-order accuracy.

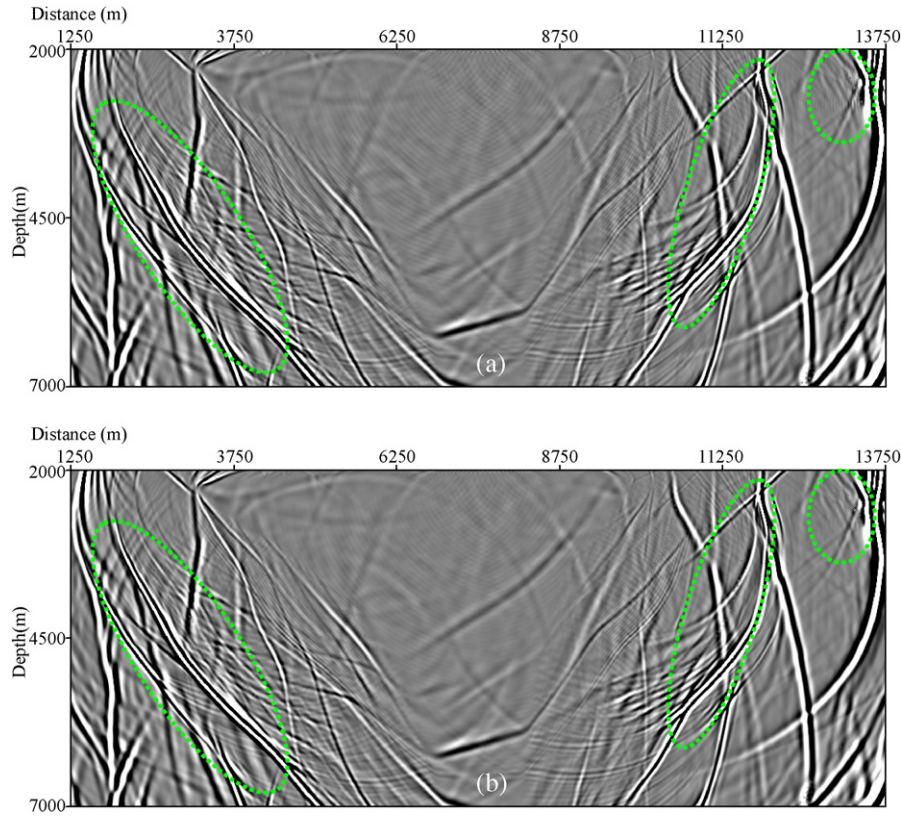


Fig. 13. Snapshots computed by the LS-based RSFD scheme: (a) the 10th-order accuracy, (b) the 18th-order accuracy.

model, the wavefield is relatively complex. However, the differences in dispersion phenomenon between the low-order and high-order schemes, the conventional and optimal schemes are obvious. When comparing the modeling results of the 10th-order and 18th-order RSFD schemes for the same method, we can see that in the same simulated region the higher-order schemes get a capacity of suppressing dispersion more effectively, and improve the numerical accuracy. In addition, this improvement is more obvious for the TE-based RSFD scheme in Figs. 11 and 14, because the artifacts in the snapshots and the seismograms by the low-order SA-based and LS-based RSFD schemes in Figs. 12(a), 13(a), 15(a), and 16(a) have been already

suppressed to a large extent. On the other hand, when we compare Figs. 11(a), 12(a), and 13(a), Figs. 11(b), 12(b), and 13(b), Figs. 14(a), 15(a), and 16(a) or Figs. 14(b), 15(b), and 16(b), we can see that under the same operator length, the SA-based RSFD and LS-based RSFD schemes have greater accuracy than the TE-based RSFD scheme. When considering a certain region in the snapshots and seismograms, we can see that the artifacts introduced by the 10th-order TE-based RSFD scheme are extremely serious. The 18th-order TE-based RSFD scheme improves the numerical accuracy of the corresponding place, but still introduces some dispersion. The 10th-order SA-based and LS-based RSFD schemes can lead to the similar effectiveness in suppressing

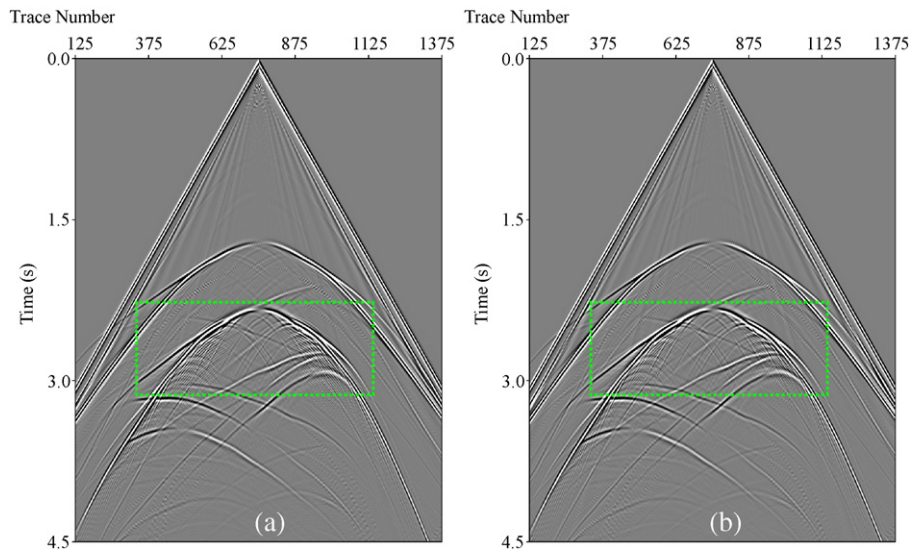


Fig. 14. Seismograms computed by the TE-based RSFD scheme: (a) the 10th-order accuracy, (b) the 18th-order accuracy.

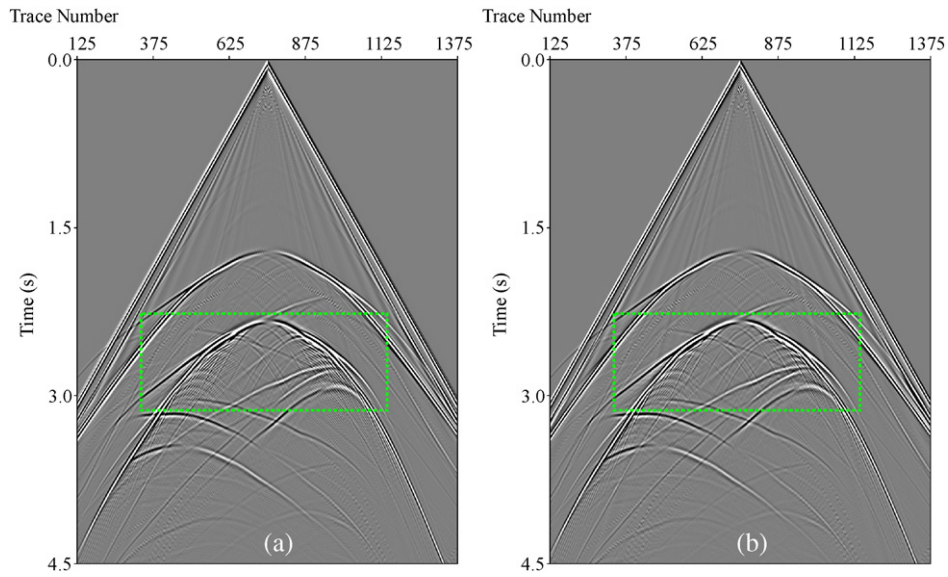


Fig. 15. Seismograms computed by the SA-based RSFD scheme: (a) the 10th-order accuracy, (b) the 18th-order accuracy.

dispersion as the 18th-order TE-based RSFD scheme does, while for the same simulated region, the 18th-order SA-based and LS-based RSFD schemes get the greatest modeling accuracy, and achieve excellent modeling results. These modeling results demonstrate that the SA-based RSFD scheme and the LS-based RSFD scheme have greater accuracy compared with the TE-based RSFD scheme, which can implement high-accuracy numerical modeling while saving operator length.

6. Conclusions

We have proposed the optimal RSFD schemes based on the SA and LS methods to achieve high-accuracy modeling results for TTI media. The theory of deriving the SA-based RSFD and LS-based RSFD coefficients is simple, and can be directly applied to the solution of the first-order spatial derivatives on rotated staggered-grid. On one hand, we make numerical analysis comparisons among the TE-based RSFD scheme, the SA-based RSFD scheme and the LS-based RSFD scheme, which demonstrate that these two proposed optimal schemes have the advantage of widening the wavenumber range with great accuracy compared with the TE-based RSFD scheme. On the other hand, we also

make further contrasts between the two optimal schemes, which show that the SA-based RSFD achieves greater accuracy at small wavenumbers, while the LS-based RSFD performs better at large wavenumbers. The numerical modeling results also provide solid evidence that the new methods outperform the conventional TE-based RSFD scheme. Meanwhile, the implementation of the SA-based RSFD and LS-based RSFD schemes is easy. If we have already got the TE-based RSFD code, we can simply exchange the existing function of computing the coefficients with that using the optimal schemes. Additionally, although our discussion of the new schemes is in 2D, they can be easily extended to 3D numerical modeling.

Acknowledgment

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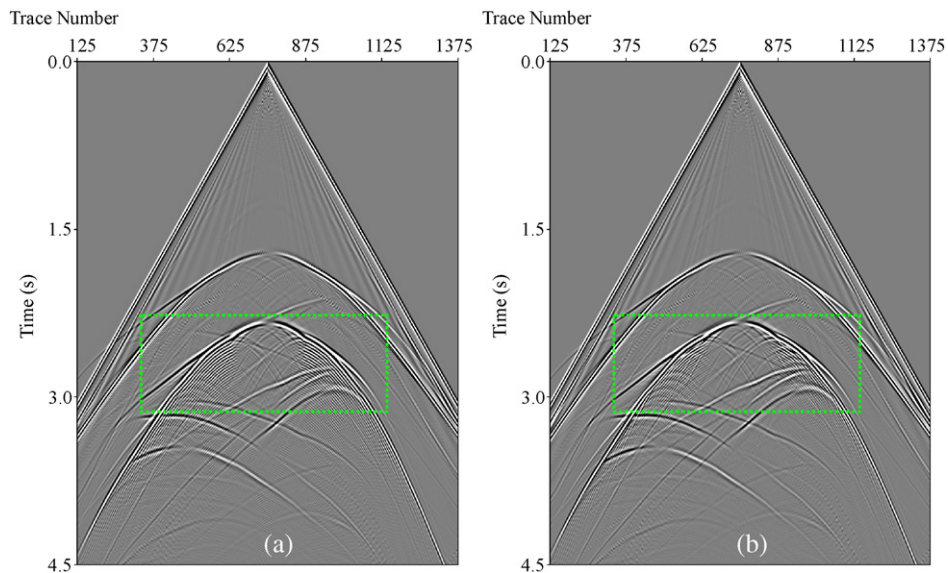


Fig. 16. Seismograms computed by the LS-based RSFD scheme: (a) the 10th-order accuracy, (b) the 18th-order accuracy.

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